

Session 1 (ASK): Exercise sheet

Foundations of Clinical Research · Level 1 · Neudata · #ClearDataClearImpact · Week 1 · Saturday

Name / group: _____ Date: _____

- > **How to use this sheet.** You do not need a computer today. Write on the sheet.
- > - Exercise 1 (warm-up): spot the PICO, on your own (3 minutes, segment 1.2).
- > - Exercises 1–2: in **pairs**, segment 1.2 (11 minutes).
- > - Exercise 3: aim, objectives and hypothesis, on your own (segment 1.3, 7 minutes).
- > - Exercise 4: your **group's Proposal builder 1** (28 minutes).
- > - Exercise 5: homework for next Saturday (synonym list).
- >
- > The answers are discussed in the room. There is no mark.

The running example (for reference)

P: Population	Adults (18+) with septic shock in the ICU
I: Intervention	Restrictive fluid strategy
C: Comparison	Usual care (standard fluids)
O: Outcome	Death from any cause
T: Time	Within 28 days

Question: *In adults with septic shock in the ICU, does a restrictive fluid strategy, compared with usual care, reduce 28-day mortality?*

Exercise 1: From clinical scenario to PICO(T) (warm-up on your own 3 minutes, then pairs 8 minutes)

Warm-up: spot the PICO (on your own, 3 minutes)

"In adults aged 65+ after hip-fracture surgery, does a daily non-drug bundle, compared with usual care, reduce delirium in the first 7 days?"

P	I	C	O	T

Pair task. Pick **ONE** scenario. Fill the PICO(T) boxes, then write the question as **one sentence**.

Tips

- For an observational question, I becomes **E** (exposure), giving "PECO".
- A descriptive question (how many? how often?) may only need **P** and **O**.
- Choose **one** primary outcome and say **when** it is measured.

Scenario A: Ventilator-associated pneumonia. Your ICU raises the head of the bed to 30–45° for ventilated patients, but compliance is patchy at night. A colleague wonders whether a twice-daily nurse checklist would reduce ventilator-associated pneumonia.

Scenario B: Pressure injuries. Several patients in your medical ward developed pressure injuries last quarter. Some staff turn patients every 2 hours, others every 4 hours. Nobody knows how often pressure injuries occur in your unit.

Scenario C: Delirium. Older patients after hip-fracture surgery often become confused. The geriatric team suggests a daily non-drug bundle (glasses and hearing aids on, orientation, sleep protection, early mobilisation).

Scenario D: Stroke door-to-needle time. Your emergency department wants to know whether a pre-hospital alert from the ambulance shortens the time from arrival to thrombolysis.

Scenario E: Medication errors. The pharmacy notices many errors when patients move from ICU to the ward. They propose a pharmacist-led medication reconciliation at transfer.

	My scenario: ____
P: who?	
I / E: what is given or what are they exposed to?	
C: compared with what?	
O: which outcome?	
T: measured when?	

My question in one sentence:

Type of question (tick one): ☐ descriptive ☐ analytical (observational) ☐ analytical (experimental)

Exercise 2: FINER traffic light (*pairs, part of the same 8 minutes; Part B is extra practice*)

Part A. Rate *your* question from Exercise 1. Colour or write **G** (green: fine), **A** (amber: needs thought) or **R** (red: must change).

Letter	Test	G / A / R	Why?
F: Feasible	Enough patients, time, money, skills, data?		
I: Interesting	To you, your team and readers?		
N: Novel	Confirms, refutes or extends what is known?		
E: Ethical	Could an ethics committee approve it?		

Letter	Test	G / A / R	Why?
R : Relevant	Could the answer change practice, policy or research?		

If any letter is **red**, rewrite the question here:

Part B (if time). Which FINER letter is the main problem in each question below?

1. "In all hospitalised adults in our country, does vitamin D supplementation reduce mortality over 10 years?" Main problem: _____
 2. "In ICU patients with septic shock, does withholding antibiotics for 24 hours reduce resistance?" Main problem: _____
 3. "Do nurses in our ICU wash their hands?" (already measured monthly by the infection-control audit for 5 years) Main problem: _____
 4. "In adults with community-acquired pneumonia, does the colour of the drug chart affect 28-day mortality?" Main problem: _____
-

Exercise 3: Aim, objectives and hypothesis (*individual, 7 minutes, segment 1.3*)

Use the question from your pair task (or the delirium warm-up question).

Aim (one sentence, starting "To compare...", "To describe..." or "To estimate..."):

Objectives (2, each with a measurable verb such as compare, estimate, describe, determine or measure; tick the primary one):

1. ☐ _____
2. ☐ _____

Vague-verb check: rewrite "To understand pressure injuries on our ward" as a measurable objective:

Hypothesis. Write **H0** (null: no difference) and **H1** (alternative: a difference) for your primary objective.

If your question is **descriptive**, write "No hypothesis (descriptive)" and say what you will *estimate* instead.

H0: _____

H1: _____

Practice: write H0 and H1 for:

a) "In ventilated ICU adults, does early mobilisation, compared with usual care, shorten ICU length of stay?"

H0: _____

H1: _____

b) "What proportion of our septic shock patients have a positive cumulative fluid balance on day 3?"

H0 / H1 needed? ☐ yes ☐ no. Because: _____

Exercise 4: Proposal builder 1 (group, 28 minutes)

Write your answers into the **Research Proposal template**

(Templates/Research_Proposal_Template.docx), Part 1. This sheet is your checklist. The search, references, problem statement and working title come **next Saturday** (Session 2, Proposal builder 2).

Suggested roles: chair · scribe (types into the template) · timekeeper · "devil's advocate" (checks FINER).

Time	Task	Done
0–7 min	1. Choose ONE problem from your unit: list 3 ideas, vote, pick one	☐
7–14 min	2. PICO(T) table and the research question in one sentence	☐
14–19 min	3. FINER check: traffic light for every letter; rewrite the question if any letter is red	☐
	4. Research, audit or QI? Write the approval route (ethics committee or local approval)	☐
19–25 min	5. Aim (one sentence), 2–3 objectives (measurable verbs; mark the primary), H0 / H1 (or "descriptive: no hypothesis")	☐
25–28 min	One group reads its question and aim aloud; the room suggests one improvement	☐

Our question: _____

FINER: F ☐ I ☐ N ☐ E ☐ R ☐ (G / A / R) **Type:** ☐ research ☐ audit ☐ QI

Quality checks before you stop

- ☐ Every PICO box is filled, and there is only **one** primary outcome with a time point.
- ☐ The question, aim, primary objective and hypothesis all talk about the **same** outcome.
- ☐ No FINER letter is red.
- ☐ You know whether it is research, audit or QI, and who must approve it.

Exercise 5: Homework (synonyms for next Saturday's search) (individual, about 15 minutes)

PubMed searches in **English**. For each PICO word in your group's question, write 2–3 English words or phrases that authors might use.

PICO element	Our word	English synonyms (2–3)
P		
I / E		
O		

Example (pressure injury): pressure ulcer · pressure injury · bedsore · decubitus ulcer.

One local number that shows the problem (an audit result, a count, an incident):

Exit question (on a sticky note, no names)

1. Which FINER letter did our running example fail, and why?
2. Rewrite as an objective: "To understand pressure injuries on our ward".

Homework before next Saturday (Session 2)

- **Group:** finish the PICO, question, aim and objectives if not done.
- **Everyone:** Exercise 5 (synonym list) and one local number that shows the problem.
- **Optional:** install Zotero (free, [zotero.org](https://www.zotero.org)) on your laptop.
- **Bring:** your group's PICO question, your synonym list, a laptop or phone per group.